MIL-H-4081B
AMENDMENT 3
23 March 1979
SUPERSEDING
AMENDMENT 2
2 April 1974

MILITARY SPECIFICATION

HOLDERS: COUNTERSINK, ADJUSTABLE, MICROMETER STOP AND CUTTERS, COUNTERSINK

This amendment forms a part of Military Specification MIL-H-4081B, 21 August 1969, and is mandatory for use by all Departments and Agencies of the Department of Defense.

PAGE 1

In paragraph 1.2:

1. After classes of cutters, add the following:

"Pilots:

Style A	With integral pilot.
Style B	With detachable drill for pilot (for cutter size see Table I)."

2. Delete the table in its entirety that shows "Size", "Body Diameter", and "Pilot Diameter".

PAGE 2

In paragraph 2.1:

1. Under "SPECIFICATIONS - Federal", add:

"PPP-P-40 Packaging and Packing of Hand Tools."

2. Under "SPECIFICATIONS - Military", delete "MIL-H-15424" along with its title.

3. Under "STANDARDS - Federal", add:

"FED-STD-H28 Screw-Thread Standards for Federal Services."

4. Under "STANDARDS - Military", add:

"MS 15444 Drills, Twist, High Speed Steel, Straight Round Shank, Short Length."

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016 by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5133

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5. After "STANDARDS - Military", add the following:

"Commercial

ANSI B46.1	Surface Texture.
ASTM E18	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Test For."

Delete paragraph 2.2 in its entirety.

PAGE 3

Delete paragraph 3.3 in its entirety and insert the following:

"3.3 Threads. All threads shall conform to class 2 tolerance as defined in FED-STD-H28."

In the third line of paragraph 3.4.2, delete "USAS B46.1" and insert "ANSI B46.1".

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Delete paragraph 3.5.1.2 in its entirety and insert the following:

"3.5.1.2 Spindle for type II holder. Unless otherwise specified (see 6.2), the cutter mounting end of the spindle shall be threaded 1/4-28UNF-2B for a length of 1/2 inch. Cutter centering angle at the thread entrance shall be 118 degrees ± 1/2 degree included angle by 0.340 inch diameter. Wrenching flats shall be provided for installing and removing cutters (see figure 2)."

At the end of paragraph 3.6, add the following:

"Configuration and dimensional requirements shall be as shown in figures 3 or 4 (as applicable) and as specified in Table I for combination drill and countersink using Style A pilots. Countersink cutters using Style B pilots may deviate from pilots sizes shown in Table I but shall have provisions for securing the pilots in place."

After paragraph 3.6.2, add paragraphs as follows:

"3.6.3 Pilots. The style of pilot shall be as specified (see 6.2).

3.6.3.1 Style A pilot. Style A pilot shall be of integral construction with its countersink cutter.

3.6.3.2 Style B pilot. Style B pilot shall be a separate drill in accordance with MS15444 and shall be attachable onto its applicable countersink cutter."

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Delete Table I in its entirety and substitute the following:

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TABLE I
Dimensional Requirements

Size	A - Pilot Exposed Length (Inches) ± 1/64	B - Body Diam. (Inches) ± .005	D - Pilot Diam. (Inches) + .000 - .001	C - Cutter angle ± 0° 30'
A	3/32	.437	.093	60°, 78°, 82°, 100°, 110°, 115°
B	1/8	.437	.125	
C	5/32	.437	.156	
D	3/16	.437	.187	
E	1/8	.500	.125	
F	5/16	.500	.250	
G	5/16	.625	.312	
H	3/32	.375	.093	
I	1/8	.375	.125	
J	5/32	.375	.156	
K	3/16	.375	.187	
L	1/4	.500	.156	
M	1/4	.875	.250	
N	1/4	.875	.312	
O	5/16	.875	.375	

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After paragraph 3.7, add the following paragraph:

"3.8 Reclaimed materials. The assemblies shall contain reclaimed materials to the maximum extent possible without jeopardizing the material quality or performance of the equipment. The reclaimed material shall be reprocessed, remanufactured, or recycled in a manner which restores them to the same chemical composition and physical properties as the materials originally selected for use on the assemblies. Reclaimed materials shall be inclusive of alloying elements applicable that have been collected from discarded solid liquid, or gaseous waste from garbage. refuse, sludge and from other collections of materials."

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On the third line of paragraph 5.1, delete "MIL-H-15424" and insert "PPP-P-40".

Entirely delete paragraph 6.2 along with its subparagraphs and insert the following:

"6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Type holder required (see 1.2).

c. Countersink cutter type, class, style, size, and countersink angle required (see 1.2, 3.6, 3.6.3, and Table I).

d. Threads, if different (see 3.5.1.2).

e. Selection of applicable levels of preservation, packaging, packing, and marking (see 5.1)."

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Add the following Table :

TABLE III
Metric Equivalents

Dimensions:			Finish		
Inches		MM	Microinch (U In.)		Micrometer (UM)
	0.001	0.03	32		0.80
	0.003	0.08	125		3.18
	0.005	0.12	Angles		
	0.006	0.15	Degrees	Radians	
1/64	0.015625	0.39	1/2 (30')	0.008727	
1/32	0.03125	0.79	5	0.087266	$\pi/36$
1/16	0.0625	1.59	45	0.785398	$\pi/4$
	0.093	2.36	60	1.047198	$\pi/3$
3/32	0.09375	2.38	78	1.361357	
1/8	0.125	3.18	82	1.43117	
	0.129	3.28	100	1.745329	$5\pi/9$
	0.156	3.96	110	1.91986	$11\pi/18$
5/32	0.15625	3.97	115	2.007129	
	0.187	4.75	118	2.059489	
3/16	0.1875	4.76			
1/4	0.250	6.35			
9/32	0.28125	7.14			
	0.312	7.92			
5/16	0.3125	7.94			
	0.340	8.64			

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(table con't)

3/8	0.375	9.52
	0.406	10.31
	0.437	11.1
1/2	0.500	12.7
9/16	0.5625	14.29
	0.600	15.24
5/8	0.625	15.88
21/32	0.65625	16.67
3/4	0.750	19.05
7/8	0.875	22.22

Custodians:
Air Force - 99
Army - GL
DSA-IP

Reviewers:
Air Force - 99
Army - GL

User Activities:
Army - AV, EL, WC

Preparing Activity:
Air Force - 99

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